CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 92-036

WASTE DISCHARGE REQUIREMENTS
AND
WATER RECLAMATION REQUIREMENTS
FOR

SALIDA SANITARY DISTRICT
AND
MR. DERK VAN KONYNENBURG
STANISLAUS COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Vail Engineering Corporation submitted a Report of Waste Discharge, dated 22 October 1991, and a site evaluation report, dated October 1991, for Salida Sanitary District and Mr. Derk Van Konynenburg (hereafter Discharger) for waste discharge requirements and Water reclamation requirements.

2. Waste Discharge Requirements Order No. 89-236, adopted by the Board on 8 December 1989, prescribes requirements for discharge of 1.2 MGD and an ultimate average flow of 2.4 MGD of treated domestic wastewater to percolation and evaporation ponds on the south bank of the Stanislaus River and to rapid infiltration basins farther from the river, about one-half mile east of the Highway 99 bridge.

3. The Report of Waste Discharge describes the treatment process and new rapid infiltration basin (RIB) disposal area at the new treatment plant just completed. The treatment process is an extended aeration sequencing batch reactor system designed to produce an effluent with 20 mg/l BOD, 20 mg/l TSS, and 10 mg/l NH3-N. The new RIB disposal area is southeast of the existing disposal ponds, further from the Stanislaus River, and about ten feet higher in elevation. The new plant recently completed (Phase 1) has an average daily dry weather flow of 1.2 MGD with plans to expand to 2.4 MGD (Phase 2) with future development.

4. Mr. Derk Van Konynenburg, the landowner adjacent to the plant, has recently entered into an agreement with Salida Sanitary District to take all the plant's treated effluent to supplement his agricultural irrigation needs; any excess wastewater will be disposed of in the rapid infiltration basins.

5. A transfer pump station (secondary transfer pump station) was constructed at the lower disposal ponds to flood irrigate Mr. Derk Van Konynenburg's lower 70-acre walnut orchard in case the plant's disposal capacity was exceeded. Adjacent to the treatment plant are the landowner's orchards which consist of 378 acres of peaches, 106 acres of

almonds and 91 acres of walnuts (Attachment C). These crops are currently irrigated by sprinkler or drip irrigation. The landowner's intention, however, is to convert all the peach orchards to drip irrigation by October 1992.

6. Effluent will be conveyed to the orchards by means of the secondary transfer pump station and a new pump station to be referred to as the main transfer pump station. The main pump station will withdraw reclaimed water from one of the RIB's and discharge it through a sand filter system into an irrigation stand pipe, and then by gravity flow into the landowners irrigation system (Attachment C). A backflow prevention structure will be constructed about 420 feet north of the MID main canal to keep reclaimed water from entering the canal. The structure will be equipped with a continuously running motorized trash screen which will keep debris from entering the flap gate.

7. Regional Board staff, by letter of 7 March 1990, allowed irrigation of the lower 70-acre walnut orchard as an interim method of disposal of excess wastewater prior to completion of expanded wastewater treatment and disposal facilities during the summer 1991. Berms were constructed to prevent wastewater runoff from the orchard. The orchard was posted and field workers informed of the use of non-disinfected wastewater on the orchard. Due to the nature of harvesting and processing of walnuts, there was no chance of contamination of the walnuts with wastewater.

8. Lands surrounding the existing disposal area are currently orchard.

9. Order No. 89-236 is neither adequate nor consistent with current plans and policies of the Board.

10. The wastewater treatment and disposal facilities, and the proposed wastewater reclamation area are in Section 28, T2S, R8E, MDB&M, with surface water drainage to the Stanislaus River, as shown in Attachment A, which is attached hereto and part of the Order by reference.

11. The Board adopted a Water Quality Control Plan, Second Edition, for the San Joaquin Basin (5C) (hereafter Basin Plan), which contains water quality objectives for all waters of the Basin. These requirements implement the Basin Plan.

12. The beneficial uses of the Stanislaus River are municipal, industrial, and agricultural supply; recreation; esthetic enjoyment; hydropower generation; and preservation and enhancement of fish, wildlife, and other aquatic resources.

13. The beneficial uses of underlying ground water are municipal, industrial, and agricultural supply.

14. The Basin Plan encourages reclamation.

15. Stanislaus County has certified a final environmental impact report (EIR) in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines. The project as approved will not have a significant effect on water quality.

16. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.

17. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 89-236 is rescinded and Salida Sanitary District and Mr. Derk Van Konynenburg, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.

2. Bypass or overflow of untreated or partially treated waste is prohibited.

3. Discharge of waste classified as 'hazardous' or 'designated', as defined in Sections 2521(a) and 2522(a) of Chapter 15, is prohibited.

4. After the new RIBs are completed and operating, the discharge of wastes to within 100 feet of surface waters shall be prohibited.

B. Discharge Specifications:

1. Neither the treatment nor the discharge shall cause a nuisance or condition of pollution as defined by the California Water Code, Section 13050.

2. The monthly average dry weather discharge flow shall not exceed 2.4 million gallons/day.

3. The discharge shall remain within the designated disposal area at all times.

4. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment, disposal and reclamation areas.

5. As a means of discerning compliance with Discharge Specification No. 4, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l. This specification does not apply to RIBs.

6. The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.

7. Wastewater shall not stand continuously in any RIB for more than 72 hours after discharge to the RIB has ceased, or the basin will be considered a holding pond.

8. Ponds shall be managed to prevent breeding of mosquitos. In particular,

    a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.

    b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.

    c. Dead algae, vegetation, and debris shall not accumulate on the water surface.

9. Public contact with wastewater shall be precluded through such means as fences, signs, and other acceptable alternatives.

10. Ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the nonirrigation season. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet (measured vertically).

11. On or about 1 October of each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification 10.

C. **Discharge Specifications: Reclamation (Agricultural Irrigation)**

1. Reclaimed wastewater may be used for irrigation of peaches, walnuts and almonds; public contact with wastewater is prohibited.

2. Uses of reclaimed wastewater must comply with the appropriate provisions of Title 22, Division 4, CCR, and specifications and requirements in this Order.

3. Peaches cannot be <u>spray irrigated</u> with reclaimed water when fruit is on the tree.

4. Peaches may be <u>spray irrigated</u> between the end of harvest and the start of fruit production (about 1 May).

5. Peaches may be <u>surface or drip irrigated</u> all year with reclaimed water provided that no fruit is harvested that has come in contact with the ground.

6. Walnuts and almonds may be <u>spray irrigated</u> under Section 60307 of Title 22 provided that irrigation by reclaimed water ceases a minimum of four (4) weeks prior to harvest.

7. Walnuts and almonds may be <u>surface</u> or <u>drip irrigated</u> under Section 60305b of Title 22 provided that irrigation by reclaimed water ceases a minimum of four (4) weeks prior to harvest.

8. A backflow prevention structure shall be constructed and maintained in good operating order, to keep reclaimed water from entering MID Main Canal.

9. To avoid the possibility of reclaimed water entering the well casing of Well No. 1 and therefore contaminating groundwater, the discharge line from the well which enters the irrigation box from the side shall be raised above the high water level of the MID Main Canal.

10. Berms shall be constructed and maintained to assure that there is no irrigation or stormwater runoff from orchards to surface water or adjacent property.

11. Nutrient content of the wastewater shall be evaluated and adjustment of the fertilizer application rates made if appropriate to meet the fertilizer needs of the orchards.

12. The District shall maintain in good working order and property operate as efficiently as possible, any facility or control system installed or used by the Discharger to achieve compliance with waste discharge requirements.

D. Sludge Disposal:

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Chapter 15 and approved by the Executive Officer.

2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer at least 90 days in advance of the change.

3. Use and disposal of sewage sludge shall comply with existing federal and state laws and regulations, and with Clean Water Act (CWA) Section 405(d) technical standards when promulgated.

E. Ground Water Limitations:

The discharge, in combination with other sources, shall not cause underlying ground water to:

1. Be degraded.

2. Contain chemicals, heavy metals, or trace elements in concentrations that adversely affect beneficial uses or exceed maximum contaminant levels specified in 22 CCR, Division 4, Chapter 15.

3. Exceed a most probable number of total coliform organisms of 2.2/100 ml over any seven-day period.

4. Exceed concentrations of radionuclides specified in 22 CCR, Division 4, Chapter 15.

5. Contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.

6. Contain concentrations of chemical constituents in amounts that adversely affect agricultural use.

F. Provisions:

1. The Discharger shall comply with the Monitoring and Reporting Program No. 92-036, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.

2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."

3. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

4. At least **90 days** prior to termination or expiration of any lease, contract, or agreement involving disposal or reclamation areas or off-site reuse of effluent, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.

5. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.

6. Mr. Derk Van Konynenburg, as owner of the real property at which the reclamation will occur, is ultimately responsible for ensuring compliance with these requirements as they relate to the use of reclaimed water on his property. Salida Sanitary District retains primary responsibility for compliance with these requirements, including day-to-day operations and monitoring. Enforcement actions will be taken against Mr. Derk Van Konynenburg only in the event that enforcement actions against Salida Sanitary District are ineffective or would be futile, or that enforcement is necessary to protect public health or the environment. Salida Sanitary District is solely responsible for the wastewater treatment plant and the RIBs.

7. The Salida Sanitary District shall provide certified wastewater treatment plant operators in accordance with regulations adopted by the State Water Resources Control Board.

8. The Discharger shall report promptly to the Board any material change or proposed change in the character, location, or volume of the discharge, or proposed change in wastewater reclamation use.

9. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.

10. If reclaimed water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes". Other uses of reclaimed water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with 22 CCR, Division 4.

11. The Board will review this Order periodically and will revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 28 February 1992.

William H. Crooks

WILLIAM H. CROOKS, Executive Officer

2/5/92/SPD:dlk

STATE OF CALIFORNIA — ENVIRONMENTAL PROTECTION AGENCY PETE WILSON, Governor

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD —
CENTRAL VALLEY REGION
13 ROUTIER ROAD, SUITE A
CRAMENTO, CA 95827-3098
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: (916) 255-3015

FILE

7 May 1993

Mr. Kent Mitchell, President
Board of Directors
Salida Sanitary District
P.O. Box 445
Salida, CA 95368

REVISED MONITORING PROGRAM FOR SALIDA SANITARY DISTRICT, STANISLAUS COUNTY

Enclosed is a revised monitoring program for Salida Sanitary District. Please implement the revised monitoring program on the effective date of this Order. The District requested revision of the monitoring program in the 1992 Annual Summary Report of Salida, transmitted by letter of 27 January 1993 from Terry Hegel.

After reviewing the 1992 annual report, the following changes were included in the revised monitoring program. Frequency of Stanislaus River monitoring was changed from monthly to every two months when using ponds closest to the river (when using ponds 2 or 3). Nitrate, total nitrogen, total dissolved solids and conductivity were deleted from Stanislaus River monitoring. No impact was measured on these constituents in 1992. Frequency of ground water monitoring was changed from quarterly to every four months. BOD_5 and total nitrogen were deleted from groundwater monitoring as these constituents are not needed.

If you have any questions, please call Sterling Davis at (916) 255-3062.

KENNETH D. LANDAU
Senior Engineer

cc+encl: Division of Water Quality, State Water Resources Control Board, Sacramento
Stanislaus County Environmental Health Department, Modesto
Mr. Terry Hegel, Modesto
Mr. Michael Gilton, P.E., Modesto
Mr. Derk Van Konynenburg, Modesto

APPROVED
author S. Davis
senior KDL

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

REVISED MONITORING AND REPORTING PROGRAM NO. 92-036

FOR
SALIDA SANITARY DISTRICT
STANISLAUS COUNTY

Specific sample station locations shall be established under direction of the Board's staff and a description of the stations shall be attached to this Order.

EFFLUENT MONITORING - DISCHARGE TO RAPID INFILTRATION BASINS

Effluent samples shall be collected just prior to discharge to the disposal facility. Effluent samples should be representative of the volume and nature of the discharge. Samples collected from the outlet structure of ponds will be considered adequately composited. Time of collection of a grab sample shall be recorded. The following shall constitute the effluent monitoring program:

Constituents	Units	Type of Sample	Sampling Frequency
20°C BOD_5	mg/l	Grab	Weekly
Suspended Matter	mg/l	Grab	Weekly
Settleable Matter	ml/l	Grab	Weekly
Total Dissolved Solids	mg/l	Grab	Monthly
Specific Conductivity	µmhos/cm	Grab	Monthly
Standard Minerals	mg/l	Grab	Annually
pH	pH Units	Grab	Weekly
Flow to upper disposal areas	MGD	Flow Meter	Daily
Flow to lower disposal areas	MGD	Flow Meter	Daily
Nitrate	mg/l as N	Grab	Monthly
Total Nitrogen	mg/l as N	Grab	Monthly

POND MONITORING

The following shall constitute the pond monitoring program: Pond monitoring shall include ponds other than RIBs.

Constituents	Units	Type of Sample	Sampling Frequency	Location of Sample
Dissolved Oxygen	mg/l	Grab	Monthly	Each pond
Pond Freeboard	feet	Measured	Monthly	Each pond

STANISLAUS RIVER MONITORING

The Discharger shall establish at least two receiving water sampling stations on the Stanislaus River, one upstream and one downstream from the disposal area. A proposal and time schedule for the sampling locations shall be submitted for review and approval **within 90 days** of the adoption of this monitoring and reporting program. Samples shall be taken every two months when using ponds closest to the river (when using ponds 2 or 3) and analyzed for the following:

Constituents	Units
Total Coliform Organisms	MPN/100 ml

MONITORING OF WASTEWATER BEING USED FOR IRRIGATION

The following monitoring shall be conducted when going to irrigation. Samples collected at the main and secondary transfer pump station will be considered adequately composited, and representative of wastewater being used for irrigation.

Constituents	Units	Type of Sample	Sampling Frequency
20°C BOD_5	mg/l	Grab	Weekly
Dissolved Oxygen	mg/l	Grab	Weekly
Settleable Solids	ml/l	Grab	Weekly
Total Coliform Organisms	MPN/100 ml	Grab	Monthly
Flow from main transfer pump station	MGD	Flow Meter	Daily
Flow from secondary transfer pump station	MGD	Flow Meter	Daily

A log shall be kept of waste being used for irrigation. It shall indicate the type of orchard being irrigated (peach, walnuts, almonds) and method of application (spray, surface or drip irrigation). Attention shall be given to the presence and condition of berms to retain tailwater onsite and to the presence or absence of:

a. standing water
b. sludge buildup
c. odor

Notes on field conditions shall be summarized in the monitoring report.

GROUND WATER MONITORING

Ground water monitoring shall be conducted in the monitoring wells recently constructed.

Prior to sampling, the depth to ground water shall be measured in all wells. Each monitoring well shall be purged a minimum of 3 well volumes (a well volume is the volume of water in the well prior to purging) and until measurements of pH and conductivity stabilize indicating that the water in the well is representative of water in the aquifer surrounding the well. The following shall be sampled and reported every four months for each well:

Constituents	Units	Type of Sample
Depth to Ground Water	Feet	Measured
Ground Water Elevation	Feet (MSL) and Hundredths	Calculated
pH	pH Units	pH Reading of Purge Water
Conductivity	µmhos/cm	Conductivity Reading of Purge Water
Nitrate	mg/l as N	Grab
Total Coliform Organisms	MPN/100 ml	Grab

REPORTING

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner to illustrate clearly the compliance with waste discharge requirements.

Quarterly monitoring reports shall be submitted to the Regional Board by the **20th day** of the month following the quarter.

The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported to the Board.

Upon written request of the Board, the Discharger shall submit a report to the Board by **30 January** of each year. The report shall contain both tabular and graphical summaries of the monitoring data obtained during the previous year. In addition, the Discharger shall discuss the compliance record and the corrective actions taken or planned which may be needed to bring the discharge into full compliance with the waste discharge requirements.

The Discharger shall implement the above monitoring program as of the date of this Order.

Ordered by: W.S. Johnson for
WILLIAM H. CROOKS, Executive Officer

7 May 1993
(Date)

Revised May 1993
SPD:njs

INFORMATION SHEET

SALIDA SANITARY DISTRICT AND
MR. DERK VAN KONYNENBURG
STANISLAUS COUNTY

Salida Sanitary District has recently entered into an agreement with Mr. Derk Van Konynenburg, the landowner adjacent to the plant to take all the plant's treated effluent to supplement his agricultural irrigation needs. Any excess wastewater will be disposed of in the rapid infiltration basins.

The District has recently completed a new wastewater treatment and disposal facility. The property is ten feet higher in elevation and farther from the Stanislaus River than the old disposal area.

The new treatment process is an extended aeration sequencing batch reactor system designed to produce an effluent with 20 mg/l BOD, 20 mg/l TSS, and 10 mg/l NH_3-N. The new plant recently completed (Phase 1) has an average daily dry weather flow of 1.2 MGD with capability to expand to 2.4 MGD (Phase 2) with future development.

A transfer pump station (secondary transfer pump station) was constructed at the lower disposal ponds to flood irrigate the lower 70 acre walnut orchard in case the plant's disposal capacity was exceeded. Adjacent to the treatment plant are the landowner's orchards which consist of 378 acres of peaches, 106 acres of almonds and 91 acres of walnuts (Attachment C). These crops are currently irrigated by sprinkler or drip irrigation. The landowner's intention is to convert all the peach orchards to drip irrigation by October 1992.

Effluent will be conveyed to the lower orchards by means of a pump station at the lower ponds. The main pump station at the upper facility will withdraw reclaimed water from one of the RIB's and discharge it through a sand filter system into an irrigation standpipe and then by gravity flow into the landowners irrigation system. A backflow prevention structure will be constructed to keep reclaimed water from entering MID main canal.

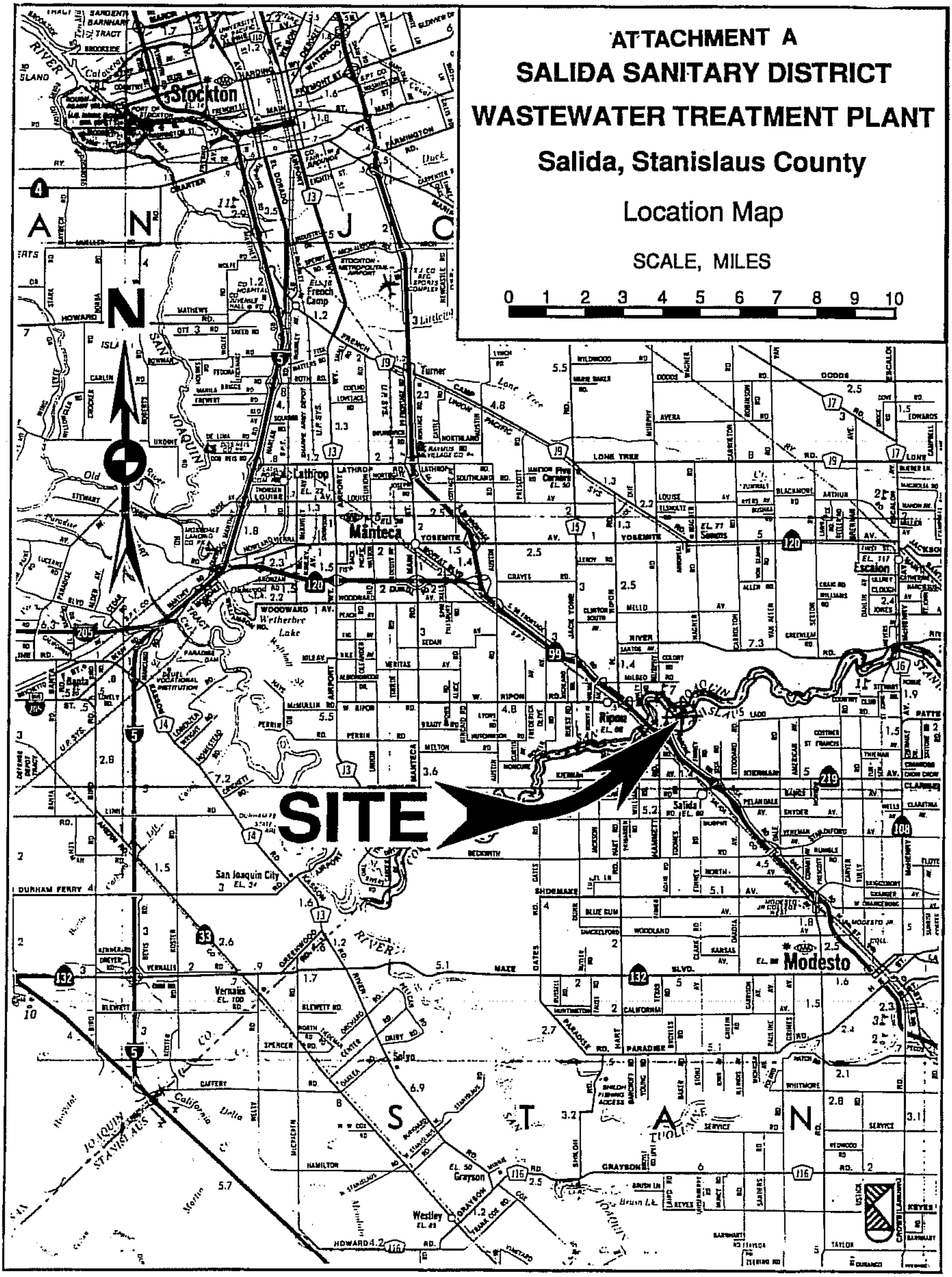
The appropriate provisions of wastewater reclamation requirements contained in Title 22, Division 4, CCR, and specifications and requirements in this Order will govern the use of wastewater for orchard irrigation. Requirements are set to assure that wastewater does not contaminate orchard crops.

ATTACHMENT A
SALIDA SANITARY DISTRICT WASTEWATER TREATMENT PLANT
Salida, Stanislaus County

Location Map

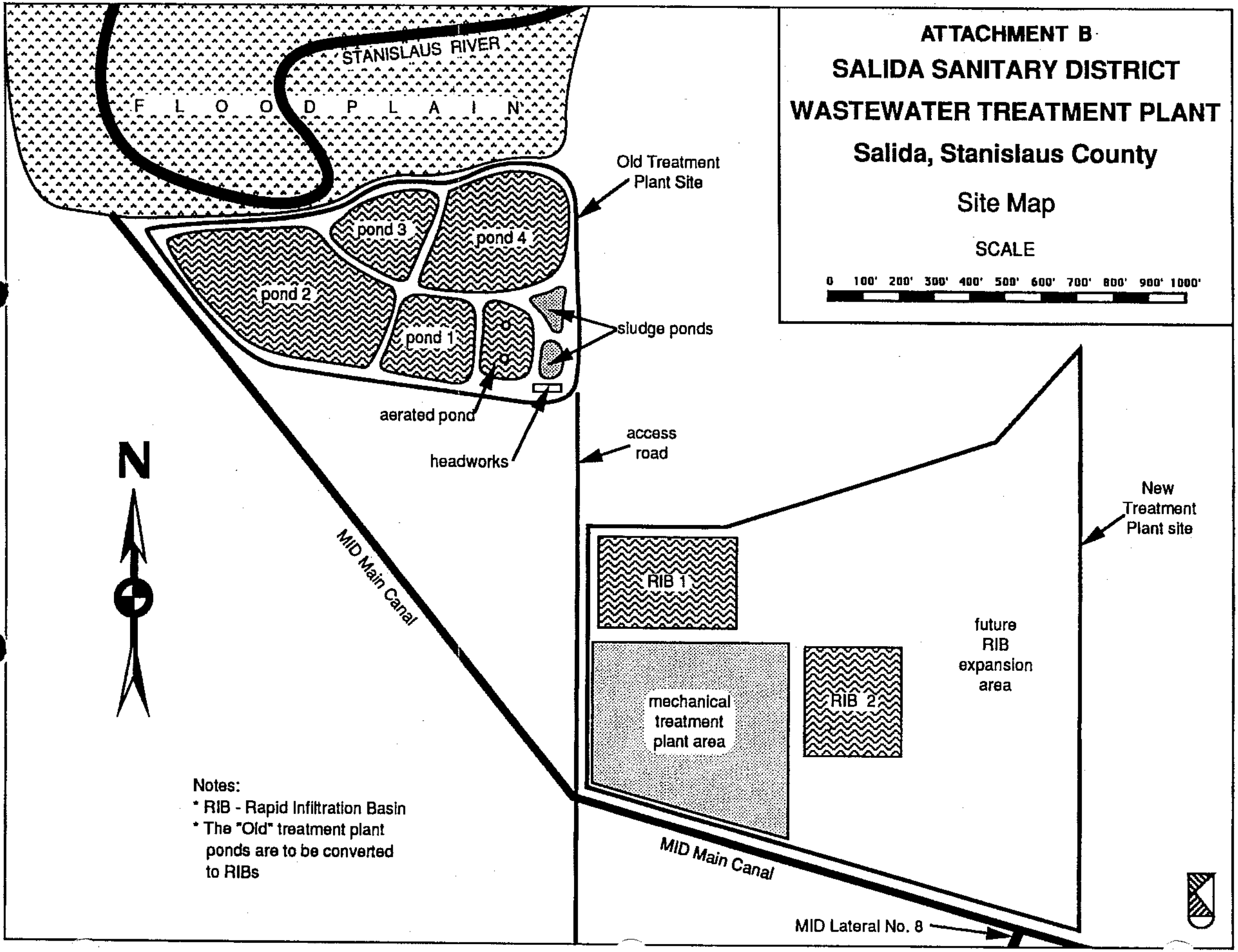
SCALE, MILES

0 1 2 3 4 5 6 7 8 9 10



ATTACHMENT B
SALIDA SANITARY DISTRICT
WASTEWATER TREATMENT PLANT
Salida, Stanislaus County

Site Map



Notes:
* RIB - Rapid Infiltration Basin
* The "Old" treatment plant ponds are to be converted to RIBs

ATTACHMENT C

SALIDA SANITARY DISTRICT

PROPOSED WASTE DISCHARGE SYSTEM

Not To Any Scale

Source: Vail Engineering Corporation, Waste Discharge Report and Application, Oct. 1991